

Integration by Substitution

ANSWER KEY



u-substitution

1 Evaluate each integral using substitution:

a $\int 2x(x^2 + 1)^5 dx$ $u = x^2 + 1$: $= \frac{(x^2 + 1)^6}{6} + C$.

b $\int \frac{x}{\sqrt{x^2 + 9}} dx$ $u = x^2 + 9$: $= \sqrt{x^2 + 9} + C$.

c $\int \cos(3x) dx$ $u = 3x$: $= \frac{1}{3} \sin(3x) + C$.

2 Evaluate $\int x^2 e^{x^3} dx$.

$u = x^3$, $du = 3x^2 dx$: $= \frac{1}{3} e^{x^3} + C$.

Definite integrals with substitution

3 Evaluate $\int_0^2 x \sqrt{x^2 + 1} dx$ exactly, changing the limits with $u = x^2 + 1$.

u goes from 1 to 5: $\frac{1}{2} \int_1^5 \sqrt{u} du = \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_1^5 = \frac{1}{3} (5\sqrt{5} - 1)$.

4 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$.

$u = \sin x$, limits 0 to 1: $\int_0^1 u^3 du = \frac{1}{4}$.

5 Evaluate $\int_1^e \frac{\ln x}{x} dx$.

$u = \ln x$, limits 0 to 1: $\int_0^1 u du = \frac{1}{2}$.

Substitution with trigonometric and exponential integrands

6 Evaluate each integral using substitution:

a $\int \sin^4 x \cos x dx$ $u = \sin x$: $= \frac{\sin^5 x}{5} + C$.

b $\int \frac{e^x}{1 + e^x} dx$ $u = 1 + e^x$: $= \ln|1 + e^x| + C$.

c $\int \sec^2 x \tan x dx$ $u = \tan x$: $= \frac{\tan^2 x}{2} + C$.

7 Evaluate $\int \frac{2x + 3}{x^2 + 3x + 1} dx$, recognizing the numerator as the derivative of the denominator.

$u = x^2 + 3x + 1$, $du = (2x + 3) dx$: $= \ln|x^2 + 3x + 1| + C$.

Choosing the substitution

8 For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3 - 1)^7 dx$ $u = x^3 - 1$, $du = 3x^2 dx$: $= \frac{1}{3} \int u^7 du = \frac{(x^3 - 1)^8}{24} + C$.

b $\int \tan x dx = \int \frac{\sin x}{\cos x} dx$ $u = \cos x$, $du = -\sin x dx$: $= -\ln|\cos x| + C = \ln|\sec x| + C$.

Substitution with radicals

9 Evaluate $\int x\sqrt{2-x} dx$ using the substitution $u = 2 - x$ (so $x = 2 - u$).

$u = 2 - x$, $x = 2 - u$, $dx = -du$:

$$\int (2-u)\sqrt{u}(-du) = \int (u-2)\sqrt{u} du = \int (u^{3/2} - 2u^{1/2}) du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C = \frac{2}{5}(2-x)^{5/2} - \frac{4}{3}(2-x)^{3/2} + C$$

10 Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} dx$ using $u = x + 1$ (so $x = u - 1$), changing the limits.

u goes from 1 to 4:

$$\int_1^4 \frac{u-1}{\sqrt{u}} du = \int_1^4 (u^{1/2} - u^{-1/2}) du = \left[\frac{2}{3}u^{3/2} - 2u^{1/2} \right]_1^4 = \left(\frac{16}{3} - 4 \right) - \left(\frac{2}{3} - 2 \right) = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}.$$

Visualizing an area found by substitution

11 The graph shows $f(x) = x\sqrt{x^2 + 1}$ on $[0, 2]$. Evaluate $\int_0^2 x\sqrt{x^2 + 1} dx$ exactly using $u = x^2 + 1$, and match it to the shaded area.

u goes from 1 to 5: $\frac{1}{2} \int_1^5 \sqrt{u} du = \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_1^5 = \frac{1}{3} (5\sqrt{5} - 1) \approx 3.393$, which equals the shaded area under the curve.

Substitution with double angle and reciprocal forms

12 Evaluate $\int \sin x \cos x dx$ two ways: (a) using $u = \sin x$, and (b) using the identity $\sin x \cos x = \frac{1}{2} \sin(2x)$. Confirm both give equivalent answers.

(a) $u = \sin x$: $\int u du = \frac{\sin^2 x}{2} + C$. (b) $\int \frac{1}{2} \sin(2x) dx = -\frac{1}{4} \cos(2x) + C$. Using $\cos(2x) = 1 - 2\sin^2 x$: $-\frac{1}{4}(1 - 2\sin^2 x) + C = \frac{\sin^2 x}{2} + C'$ — the same family, differing only by the constant.

13 Evaluate $\int \frac{1}{x \ln x} dx$ using $u = \ln x$.

$u = \ln x$, $du = \frac{1}{x} dx$: $\int \frac{1}{u} du = \ln|u| + C = \ln|\ln x| + C$.

Substitution in motion and rate problems

14 A particle's velocity is $v(t) = t \sin(t^2)$ m/s. Find its displacement from $t = 0$ to $t = \sqrt{\pi}$, using $u = t^2$.

$u = t^2$, $du = 2t dt$, limits 0 to π : $\frac{1}{2} \int_0^\pi \sin u du = \frac{1}{2} [-\cos u]_0^\pi = \frac{1}{2} (1 + 1) = 1$ m.

15 Evaluate $\int \frac{(\ln x)^2}{x} dx$ using $u = \ln x$.

$u = \ln x$, $du = \frac{1}{x} dx$: $\int u^2 du = \frac{u^3}{3} + C = \frac{(\ln x)^3}{3} + C$.

Recognizing when substitution simplifies a problem

- 16** Evaluate $\int (x + 1)\sqrt{x^2 + 2x + 5} \, dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.

$$u = x^2 + 2x + 5, \, du = (2x + 2)dx = 2(x + 1)dx: \frac{1}{2} \int \sqrt{u} \, du = \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C = \frac{1}{3} (x^2 + 2x + 5)^{3/2} + C.$$